WORKED SOLUTION

STEP III 2017, Question 3

Pure • Roots of Polynomials

STEP III 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 6.4, *Solving the Quartic*: https://beyondstep.keithmcnulty.org/ch06/quartic.html

The key idea

The coefficients of any polynomial are symmetric functions of its roots, so the coefficient A of the cubic is minus the sum of its three roots, and that sum is a familiar symmetric function of α, β, γ, δ. For the specific quartic, the cubic has an integer root, so it factorises completely. Knowing $\alpha\beta + \gamma\delta$ then splits the quartic into two quadratic factors, one with roots α, β and one with roots γ, δ, and the remaining coefficients pin down those factors.

Solution

Finding A

Throughout we use Vieta's relations for the quartic:

$$\alpha+\beta+\gamma+\delta = -p, \qquad \alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta = q,$$
$$\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta = -r, \qquad \alpha\beta\gamma\delta = s.$$

The cubic is monic with roots $y_1 = \alpha\beta + \gamma\delta$, $y_2 = \alpha\gamma + \beta\delta$ and $y_3 = \alpha\delta + \beta\gamma$, so

$$y^3 + Ay^2 + \cdots = (y - y_1)(y - y_2)(y - y_3),$$

and comparing coefficients of y^2 gives $A = -(y_1 + y_2 + y_3)$. But $y_1 + y_2 + y_3$ contains each of the six products of two different roots exactly once, so it equals q. Hence

$$A = -q.$$

Part (i)

With $p = 0$, $q = 3$, $r = -6$, $s = 10$ the cubic has coefficients

$$A = -3, \qquad pr - 4s = -40, \qquad 4qs - p^2 s - r^2 = 120 - 0 - 36 = 84,$$

so it is

$$y^3 - 3y^2 - 40y + 84 = 0.$$

Any integer root divides 84. Trying small divisors, $y = 2$ gives $8 - 12 - 80 + 84 = 0$, so $y - 2$ is a factor:

$$y^3 - 3y^2 - 40y + 84 = (y - 2)(y^2 - y - 42) = (y - 2)(y - 7)(y + 6).$$

The roots are 2, 7 and -6; the largest is 7, so

$$\alpha\beta + \gamma\delta = 7.$$

Part (ii)

Expanding,

$$(\alpha + \beta)(\gamma + \delta) = \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta,$$

which is the sum of all six pair products with $\alpha\beta$ and $\gamma\delta$ removed. So

$$(\alpha + \beta)(\gamma + \delta) = q - (\alpha\beta + \gamma\delta) = 3 - 7 = -4.$$

Next, $\alpha\beta$ and $\gamma\delta$ have sum 7 and product $\alpha\beta\gamma\delta = s = 10$. They are therefore the roots of

$$t^2 - 7t + 10 = (t - 5)(t - 2) = 0,$$

and since $\alpha\beta > \gamma\delta$ we get $\alpha\beta = 5$ and $\gamma\delta = 2$.

Part (iii)

Let $u = \alpha + \beta$. Since $p = 0$, the sum of all four roots is zero, so $\gamma + \delta = -u$, and part (ii) gives $-u^2 = -4$, so $u = \pm 2$.

? **Ask yourself** Which sign of u is correct, and how can we tell?

We haven't yet used r. Grouping the four triple products in pairs,

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = 5(-u) + 2u = -3u.$$

This equals $-r = 6$, so $u = -2$. (The other sign would give $r = +6$, a different quartic.) Thus

$$\alpha + \beta = -2, \quad \alpha\beta = 5, \qquad \gamma + \delta = 2, \quad \gamma\delta = 2.$$

So α, β are the roots of $x^2 + 2x + 5 = 0$, giving $x = -1 \pm 2\mathrm{i}$, and γ, δ are the roots of $x^2 - 2x + 2 = 0$, giving $x = 1 \pm \mathrm{i}$. The quartic

$$x^4 + 3x^2 - 6x + 10 = (x^2 + 2x + 5)(x^2 - 2x + 2) = 0$$

therefore has roots

$$x = -1 + 2\mathrm{i}, \quad -1 - 2\mathrm{i}, \quad 1 + \mathrm{i}, \quad 1 - \mathrm{i}.$$

Checks

- Multiply out $(x^2+2x+5)(x^2-2x+2)$: the x^3 terms cancel and we get $x^4+3x^2-6x+10$, exactly the quartic.
- With these roots, $\alpha\gamma+\beta\delta$ and $\alpha\delta+\beta\gamma$ (in some order) should be the other two roots of the cubic, 2 and -6. Their sum is $q-7=-4=2+(-6)$, as it should be.
- The formula $A=-q$ was also confirmed by computer on a thousand quartics with randomly chosen complex roots.

Common slips

- Taking $A=q$: the y^2 coefficient is *minus* the sum of the roots.
- Stopping at $u=\pm 2$ and giving both possibilities: only one sign is consistent with $r=-6$.
- Expecting real roots: this quartic is positive for all real x, so all four roots are complex.

Beyond STEP

The cubic here is the *resolvent cubic* of the quartic, and it is the heart of Ferrari's method: solving any quartic reduces to solving a cubic, then quadratics. Why this works is explained by Galois theory, which studies how permutations of the roots act on expressions like $\alpha\beta+\gamma\delta$ (each of the three is fixed by a subgroup of eight of the 24 permutations). Allenby's *Rings, Fields and Groups* is a good place to start on these ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.